Journal of Space Safety Engineering xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of Space Safety Engineering

journal homepage: www.elsevier.com/locate/jsse

Could artificial intelligence improve astronaut debriefing processes? A review from psychological and human–computer interaction perspectives

Simone Caso*, Yannick Hill

Department of Human Movement Sciences, Faculty of Behavioural and Movement Sciences, Vrije Universiteit Amsterdam, Amsterdam, the Netherlands

ARTICLE INFO

Article history:
Received 26 June 2026
Received in revised form 9 October 2026
Accepted 10 October 2026
Available online xxx

Keywords:
Artificial intelligence
Astronaut training
Debriefing
Ecological
Large language models

ABSTRACT

Debriefing is a cornerstone of astronaut training and performance, functioning as a structured reflective practice that transforms operational experience into learning. In the extreme environments of human spaceflight, effective debriefing supports not only safety and performance improvement but also psychological adaptation under high cognitive and physical demands. However, traditional debriefing practices rely on human facilitation, subjective recall, and fragmented data sources, which can limit their consistency and analytical depth. Recent developments in artificial intelligence (AI), like large language models, provide new opportunities to enhance debriefing processes. These technologies could support automated transcription, data integration, structured feedback generation, and self-regulated learning. Drawing on ecological psychology and self-regulated learning theories, this perspective paper examines how AI may support debriefing in astronauts training, as well as the potential psychological implications of AI applications. We argue that the integration of AI may enable such precise and data-informed feedback, which could help to individualize the debriefing process. However, the integration of AI may also raise important challenges related to data privacy, human oversight, technical implementation, and ethical use in such contexts.

© 2026 The Author(s). Published by Elsevier Ltd on behalf of International Association for the Advancement of Space Safety. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/)

1. Introduction

Debriefing is widely recognized as one of the most effective learning and performance-improvement practices across high-stakes domains such as aviation, medicine, and military operations [1–4]. In human spaceflight, debriefing is used to "translate" experience from complex mission operations into actionable understandings for future training and possibly procedures [5,6]. For instance, astronauts engage in debriefings following both simulated and real missions, such as Neutral Buoyancy Laboratory sessions, high-fidelity virtual simulations, and actual extravehicular activity (EVA), to analyze the performance and related psychological factors (e.g., the processes driving the decision-making) [7–10]. Astronauts operate within what Wilson [11] termed a distributed cognitive environment, in which perception and other psychological processes, such as decision-making, are distributed across individuals, tools, and artifacts [7,12,13]. In such contexts, effective debriefing has both operational and training purposes. For instance, debriefing reinforces mission safety protocols while also fostering psychological topics like self-regulation, which lead to adaptive expertise [14–16,4]. Further, it helps the astronauts to identify the underlying psychological processes that contribute to their performance. For example, astronauts may learn to recognize which motor responses were effective, which were unsuccessful, and which require improvement under specific dynamic conditions.

In this sense, debriefing could be viewed through the lens of ecological psychology, where learning arises from re-attunement to affordances (i.e., action opportunities) in the environment [17–23,4]. Accordingly, astronauts must learn to perceive and act upon invariant structures in a variable environment that "indicate" opportunities for action or risk [7,18,13]. However, despite the complexity of modern space (training) programs, debriefing methods are largely manual and instructor-dependent [6,24]. Much of the reflective process relies on the facilitator's note-taking and subjective recall. For example, during long-duration missions, such as those aboard the International Space Station (ISS), the time and cognitive resources available for detailed debriefings are often limited. At the same time, mission complexity is expected to increase

* Corresponding author at: Faculty of Behavioural and Movement Sciences, Van der Boechorststraat 9, Amsterdam, the Netherlands.
E-mail address: simone.caso@hotmail.it (S. Caso).

https://doi.org/10.1016/j.jsse.2026.10.002
2468-8967/© 2026 The Author(s). Published by Elsevier Ltd on behalf of International Association for the Advancement of Space Safety. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/)

Please cite this article as: S. Caso and Y. Hill, Could artificial intelligence improve astronaut debriefing processes? A review from psychological and human–computer interaction perspectives, Journal of Space Safety Engineering, https://doi.org/10.1016/j.jsse.2026.10.002

further as lunar and Mars missions become a reality. Consequently, opportunities for systematic learning from experience may be lost or captured inconsistently [5].

In parallel, the emergence of artificial intelligence (AI) presents new possibilities for supporting this process. Technological advances in large language models (LLMs), natural language processing (NLP), and speech-to-text systems have transformed the capacity of machines to process human communication and extract meaningful information [6,25–28]. These tools can transcribe spoken interactions and detect the emotional tone. Combining these communicative elements with comprehensive summaries of the performance data (i.e., flight path and control inputs of a simulation), could support the generation of coherent and context-sensitive feedback, all essential factors for effective debriefings [29,30,4]. AI applications in astronautics, have already begun to emerge in mission analysis, anomaly detection, and human–machine interfaces [24,31,32]. However, their potential in supporting astronauts' learning through debriefing is still underexplored. Hence, this perspective paper reviews how AI technologies could enhance astronaut debriefing by automating documentation, improving feedback precision, and fostering reflective learning from an ecological psychology perspective [17,20] and self-regulated learning theories [33–36].

2. Debriefing in human spaceflight

Since the early days of National Aeronautics and Space Administration (NASA)'s human spaceflight program, crew debriefings have been a cornerstone of operational learning for both astronauts and ground teams**.** For instance, debriefings following Apollo missions 11 through 17 were extensive, sometimes lasting several days, as astronauts, engineers, and other professionals reviewed the performance and effectiveness of the crew, hardware, software, and mission techniques across each phase of the mission. These sessions reviewed technical performance, and in some cases, psychological factors such as decision-making, as well as issues that emerged during the (simulated) flight [37–41]. Debriefings were attended by representatives also from the astronaut office, flight operations, and crew systems, allowing these experts to contribute information and pose questions to the crew when necessary. This process provided a critical feedback loop, enabling NASA to translate experiential understandings into modifications in the spacecraft design, mission protocols, and training curricula. For example, many procedural and engineering developments across the Apollo, Space Shuttle, and the ISS eras can be traced directly to systematic post-mission debriefings [42–45].

Indeed, the Space Shuttle-era debriefing reports highlighted the importance of interdisciplinary integration. Feedback from pilots, payload specialists, and mission controllers was categorized into technical, procedural, and human factors domains, fostering collaboration between engineering and science teams. This integrative approach anticipated current perspectives on socio-technical system learning, where psychological understanding is treated as equally critical to engineering performance. Conversely, inputs gained from crew debriefings often informed the design and refinement of spacecraft systems, cockpit interfaces, and operational procedures, showing the reciprocal relationship between human performance and engineering development. Further, debriefing practices became systematized through the introduction of Flight Crew Debriefing Reports, which documented hundreds of post-flight observations per mission, including both technical and human performance data [46]. These reports were analyzed across multiple missions to detect trends in system performance, procedural compliance, and crew workload.

In parallel, research conducted at NASA Ames Research Center explored methods to enhance the cognitive and instructional dimensions of debriefing. For instance, Martin et al. [47] demonstrated that structured instructor training programs, such as the Special Purpose Operational Training debrief model, significantly improved the quality and consistency of crew reflections. By equipping instructors with behavioral observation tools and structured facilitation guides, the program increased the proportion of Crew Resource Management-related discussions and encouraged greater crew participation in identifying operational and interpersonal challenges. These findings highlighted that debriefing quality depends not only on data review but also on the psychological design of reflective dialogue, linking performance factors to cognitive and communicative behaviors. In contemporary human spaceflight programs, debriefing is equally important, though its structure and frequency have evolved. Astronauts participate in multiple phases of debriefing throughout their training cycle and mission tenure. Pre-flight debriefings occur after simulator sessions, exercises, or simulations, allowing trainees to reflect on procedural accuracy and other aspects like situational awareness. In-flight debriefings occur during the mission. For instance, operational debriefings take place after EVAs, robotic arm operations, and scientific experiments. As an example, during Increment 75, the increment team met weekly with an ESA crew member to debrief ongoing scientific activities and experiments. For instance, E4D (see [48]) was a newly developed exercise device that the ESA astronaut was commissioning, and their regular in-flight feedback was essential for the further development and optimization of the exercise protocols. Post-flight debriefings**,** conducted after return to Earth, integrate physiological, cognitive, and behavioral data with astronauts' subjective reflections to refine both individual training and mission system design [5,49].

Each type of debriefing has a different function but shares a common purpose: to close the loop between experience, feedback, and continuous improvement**.** As Dempsey and Barshi [6] emphasized, debriefings allow astronauts and mission controllers to "close the loop" between performance and learning, thereby embedding reflection within the operational rhythm of spaceflight. During a typical debriefing session, multiple data sources are integrated. Audio and video recordings from mission operations, telemetry and environmental data, task checklists, and even biometric measurements are brought together to reconstruct what "happened". Afterwards, astronauts and instructors review mission timelines and event markers, pausing to discuss decision points or deviations from expected outcomes. Facilitators, often senior astronauts, use structured questioning techniques to encourage reflective dialogue**,** focusing on what went well, what could be improved, and what was learned. This process supports both individual self-awareness and team learning, enabling the group to extract meaning from complex, time-pressured operations [49–51].

Despite its proven benefits, current debriefing practice faces several constraints. The first is time pressure. During missions, operational demands and fatigue often limit opportunities for detailed reflection, resulting in abbreviated or postponed analyses. Meanwhile, even post-flight, astronauts must balance debriefings with medical assessments and other activities (e.g., public engagements), leaving limited windows for in-depth review. The second challenge is data fragmentation. Communication logs, video footage, telemetry, and handwritten notes are stored across unrelated systems, requiring manual collation by analysts and instructors. This fragmentation makes it difficult to achieve a holistic understanding of events and increases the likelihood that subtle performance patterns go unnoticed. Thus, a comprehensive integration of these data fragments is essential for optimizing learning. A third limitation concerns human memory and bias**.** When debriefings depend on subjective recall, especially under fatigue or emotional stress, memory distortions and selective attention can affect accuracy [52]. Astronauts, like all profession-

als, are susceptible to hindsight bias or self-presentation effects, emphasizing certain details while minimizing others, particularly when performance is being evaluated. This affects not only the quality of individual reflection but also the reliability of the perceptions drawn for system improvement. Finally, debriefings are dependent on facilitator expertise. Skilled facilitators could guide reflective dialogue effectively, but this reliance introduces variability between instructors and limits scalability [2,47]. Accordingly, the integration of AI-based tools into the debriefing process could be a promising way to address these challenges.

3. AI applications and psychological implications

Recent developments in AI, particularly through LLMs and speech-to-text systems, have transformed how human communication can be captured [25,27,29,53]. LLMs, such as GPT-4 or Gemini, are trained on extensive multimodal datasets to generate text that copies human dialogue [28]. Meanwhile, speech-to-text systems, such as OpenAI's Whisper or Google Cloud Speech Recognition, can now transcribe multi-speaker conversations with accuracy, even under noisy and dynamic conditions [54]. Hence, these AI tools can automatically capture, align, and analyze performance data from multiple sources, transforming fragmented records into coherent narratives [55,56]. Consequently, AI tools could assist in archiving and indexing debriefing data (e.g., crew communication during specific maneuvers), enabling future crews and training designers to trace the evolution of specific skills or procedures over multiple missions, a capability that manual documentation rarely achieves. In real time or shortly after task execution, algorithms could also examine and highlight deviations, making these patterns visible to both astronauts and facilitators (e.g., instructors). This enhanced visibility could support more comprehensive, data-rich reflection and improve the fidelity of feedback [26,57,58,4].

These developments may be particularly important for long-duration missions, such as the Artemis operations or future Mars expeditions [59–61]. As these types of missions become longer and more complex, debriefing complexity will increase exponentially due to the accumulation of operational data and the limited bandwidth for human analysis. Furthermore, communication delays between Mars and Earth can range from approximately 4 to 24 min one-way (8 to 48 min round-trip), depending on planetary alignment, making real-time feedback from ground-based instructors impractical [62]. Hence, crews will need to perform autonomous self-debriefing and they will need to interpret their own data and regulate their learning without immediate external input. For example, there are currently free-flying robots on board the ISS (e.g., Astrobee, CIMON, and Int-Ball) that support various aspects of human spaceflight while working side-by-side with astronauts. In particular, CIMON uses AI to provide verbal assistance [63]. So, in such contexts, AI-supported systems could support structured reflection by summarizing performance metrics and suggesting adaptive strategies through conversational interfaces such as chatbots (e.g., Siri or ChatGPT) [64]. As Buckle et al. [5] argued, future crews must be capable of reflecting on and adjusting their behavior without continuous ground supervision. AI-assisted debriefing may therefore become an essential operational support mechanism, sustaining learning and reflection during isolated missions [10]. In that respect, it could provide a safe space for emotional release [65]. Indeed, such feedback sessions may mitigate evaluation anxiety and foster a climate in which individuals feel comfortable admitting mistakes and discussing challenges without fear of potential negative consequences or judgment [66].

Accordingly, AI could also be designed as an interactive reflective partner, capable of engaging astronauts in structured dialogues that stimulate self-explanation [26,30,55]. When trained on mission-specific communication patterns and task data, LLMs could simulate guided reflection by posing context-sensitive questions such as: "Which environmental signs indicated that the task was deviating from the plan?" or "How did communication timing affect team coordination with your colleagues?" These prompts could support metacognitive processes**,** helping the astronauts to articulate their reasoning and make tacit knowledge explicit, a process well-established in self-regulated learning research [36,53]. This AI-mediated reflection could occur autonomously, providing psychological continuity in environments where human assistance is limited. Hence, during communication delays, astronauts could engage in AI-assisted self-debriefing to consolidate learning while supporting human factors skills such as emotional regulation, self-monitoring, and reflective practice. This system would thus function as a metacognitive support framework, enabling continuous self-evaluation and growth even in isolation. Over time, this interaction could also foster adaptive expertise and self-efficacy,[1] key factors in performance psychology and resilience training [33,34,67,68]. Meanwhile, they could also generate personalized reflection prompts, visualize mission data to aid comprehension, and even simulate alternative scenarios for experiential learning.

These developments have also implications for self-regulation as AI may introduce applications that could augment how the astronauts externalize, observe, and interpret their own thought processes [16]. That is, by automatically capturing speech, behavior, and physiological signals, AI could generate psychologically meaningful representations of experience**,** allowing astronauts to engage in reflection that is both immediate and evidence based. Hence, instead of relying on memory, which is subject to distortion and bias [52], astronauts could review precise AI-generated accounts of what was said, when it was said, and potentially the emotional context of the interaction. Such tools could also assist psychologists and behavioral scientists in understanding how emotional dynamics evolve during such confinement and isolation [10,69], perhaps with the addition of video behavioral analysis [70]. For instance, by quantifying factors like the emotional tone and conversational reciprocity, AI systems could act as mirrors of interpersonal functioning, supporting early detection of issues such as interpersonal tension or burnout risks.

Debriefing in astronautics is therefore fundamentally an educational, training, and wellbeing process grounded in reflection, feedback, and adaptation [1,25,55,4,71] as it enables astronauts to transform operational experiences into structured learning [6]. Within the framework of self-regulated learning**,** debriefing supports the cyclical processes of forethought, performance monitoring, and self-reflection [36,72,73]. By embedding AI into these cycles, the learning process becomes richer and more individualized. Accordingly, ecological psychology may provide a framework for conceptualizing astronaut debriefing as a process of re-attunement to affordances, the action possibilities offered by the environment relative to an individual's capabilities [20,22]. Astronauts operate in environments where perception–action couplings are continuously challenged, for example, altered gravity changes proprioceptive feedback, time-delayed communications disrupt coordination, and high-stress conditions affect situational awareness [7,10,62]. Debriefing may allow the astronauts to recalibrate these couplings, to identify what environmental information was effective, which was missing, and how their actions corresponded to mission constraints. When supported by objective data (e.g., video, telemetry, and biometric metrics), this process strengthens the link between perception, action, and reflection, forming the foundation for adaptive expertise [70]. Finally, AI systems could monitor performance patterns by identifying recurrent cognitive or behavioral tenden-

[1] Self-efficacy is the belief in one's ability to successfully perform a specific task or achieve a desired outcome [99].

cies, such as hesitations during high-workload periods or patterns of communication asymmetry. Through adaptive feedback algorithms, the system could generate personalized learning plans and recommend targeted exercises. This personalized approach would align with the principles of deliberate practice and feedback-driven learning [74], and could support psychological factors such as perceived competence and confidence.

4. Challenges

Beyond its potential applications and benefits, the integration of AI tools may also introduce several challenges. For instance, from the psychological point of view, the presence of an evaluative AI system may alter communication behaviors or inhibit emotional openness, threatening psychological safety (i.e., a prerequisite for effective learning) [75,76]. Consequently, maintaining transparency about the AI system's purpose, limitations, and data-handling practices would be important for preserving trust and authenticity in debriefing sessions [77,78]. In addition, data protection and human supervision require careful attention. In this context, safeguarding privacy becomes particularly important when AI systems record, store, or analyze sensitive performance information such as spoken dialogue, behavioral indicators, or decision-making patterns. These risks may be further amplified when data are processed through cloud-based infrastructures or when access management is insufficiently controlled [16,79,80].

To further reduce the risk of misuse, transcripts could be anonymized or partially redacted to remove personally identifiable or sensitive information prior to archiving. In practice, this process may involve the removal or "masking" of personally identifiable information (e.g., names, voice identifiers, biometric markers, or references to specific individuals or mission-sensitive events) [81–83]. Such measures would be essential to protect individual privacy, particularly in small, high-visibility populations such as astronaut crews, where re-identification risks remain high even when datasets are partially anonymized. Further, anonymization is important for reducing evaluative pressure and encouraging openness during reflective dialogue. For example, if participants are assured that their verbal reflections will not be directly attributable to them, they may be more willing to express uncertainty, emotional strain, or perceived errors, elements that are critical for their development and psychological safety [84].

From the technical perspective, maintaining human oversight is also essential to ensure that algorithmic analyses are interpreted appropriately and that users do not become overly dependent on the automated feedback. Accordingly, human instructors and supervisors should validate AI-derived outputs, applying expert contextual judgment and ethical discernment that current technologies are not yet capable of providing [53,75,85,86]. Additionally, it would be important to have protocols for the management, storage, and eventual disposal of debriefing transcripts. These records would need to be safeguarded in compliance with applicable data protection frameworks, with access restricted to authorized personnel only. Given the potential vulnerabilities associated with cloud-based platforms, the use of locally hosted or hybrid storage systems may provide greater control over data access and cybersecurity measures [87,88]. Meanwhile, ongoing improvements to speech-to-text algorithms would be necessary to overcome limitations such as accent variability, speech overlap, and environmental noise [25]. Refining these systems would enhance both the accuracy and efficiency of AI-supported debriefing processes [89]. This is particularly important in contexts where English is the operational lingua franca. Hence, collaboration among technology developers, professional users, and international regulatory bodies (e.g., the Federal Aviation Administration and International Civil Aviation Organization for the aviation), is important for the safe and effective integration of LLMs and speech-to-text systems into debriefing practices. From the psychological perspective, as AI systems increasingly participate in data interpretation, psychologists would need to consider how algorithmic mediation influences what counts as evidence of learning or emotion. Questions regarding validity, such as whether AI-derived metrics truly capture human psychological constructs, will be important to the emerging "dialogue" between psychology and AI [29]. Operationally, AI-augmented debriefing could transform how training programs are structured. Automated transcription and analysis would relieve instructors of labor-intensive documentation tasks, allowing them to concentrate on coaching and emotional support. For instance, real-time feedback during simulations could highlight communication breakdowns or workload imbalances, enabling immediate corrective action. Over successive sessions, machine-learning models could chart learning trajectories for individuals and teams, providing objective indicators of readiness.

Finally, in the context of deep space missions, a learning challenge will be maintaining ecological validity, that is, ensuring that simulated training (e.g., analog studies [90,91]) preserve the perceptual and cognitive demands of real operations [92–95]. AI technologies could enhance representativeness by integrating multimodal data sources (e.g., from helmet camera footage, biometric signals, and telemetry streams) into synchronized feedback displays. For instance, an AI-enhanced debriefing dashboard could overlay voice transcripts with physiological and environmental data, allowing astronauts and instructors to visualize factors such as how oxygen consumption, communication latency, and movement trajectories co-evolve during an EVA. Psychologically, this multimodal feedback may help the learners to reconstruct the perception–action couplings that underlie expert performance, thereby supporting attunement to affordances and adaptive calibration [13,16; see also Ecological Interface Design theories, [96–98]]. Yet even here, caution regarding privacy and the use of sensitive data is important. Table 1 provides a summary of the current debriefing issues, their limitations, potential AI capabilities, and the associated operational and learning benefits.

5. Limitations and future research

In order to validate the AI applications reviewed in this paper, future research could conduct experimental and field studies to assess the validity of AI-assisted debriefing. For instance, researchers could investigate which improvements to speech-to-text algorithms are necessary to overcome limitations such as accent variability, speech overlap, and environmental noise [25]. Secondly, researchers could examine how AI-mediated debriefing affects astronauts' learning outcomes and emotional well-being. Key variables may be cognitive workload, metacognitive accuracy, trust in AI, and psychological safety. For example, comparative study designs could examine differences between human-only, AI-supported, and fully autonomous debriefing formats. Moreover, cross-disciplinary collaboration between AI developers and other experts (e.g., psychologists and human-factors engineers) is important for creating systems that are both technically and psychologically robust. Additionally, despite recent advances in LLMs, NLP, and speech-to-text technologies, their reliability in operational and spaceflight contexts is currently not studied. Future research should therefore evaluate the systematic validation of AI-assisted debriefing tools in operational and spaceflight-relevant environments. Factors such as background noise, overlapping speech, communication delays, mission-specific terminology, and cultural or individual differences may reduce system accuracy and increase the risk of false positives, false negatives, or inaccurate summaries. Consequently, future studies would need to establish clear evaluation criteria, including transcription accuracy, summary fidelity,

Table 1
Current debriefing limitations to potential AI-supported functions and operational benefits.

Current debriefing issue	Limitations in astronaut training/operations	Potential AI capability	Operational or learning benefit	Example use case
Time pressure	During missions, operational demands and fatigue can limit opportunities for detailed reflection, while post-flight schedules may also restrict the time available for in-depth review.	Automated transcription, summarization, and event indexing from voice, video, telemetry, and task data.	It could reduce the documentation on instructors and allow astronauts and facilitators to focus more on interpretation, learning, and corrective action.	After an EVA simulation or neutral buoyancy session, AI could generate a first structured summary of key events, deviations, and discussion points before the human-led debrief.
Fragmented data sources	Communication logs, video footage, telemetry, and handwritten notes may be stored across unrelated systems, making it difficult to reconstruct events holistically.	Multimodal data integration, aligning transcripts with video, telemetry, biometric data, and event markers.	It could support a more coherent reconstruction of what happened and help to identify performance patterns that may otherwise remain unnoticed.	In a robotic arm, EVA, or scientific operations debrief, AI could synchronize communication timing with task events and telemetry to support precise analysis of decision points.
Memory bias and subjective recall	When debriefing relies mainly on retrospective recall, fatigue, stress, hindsight bias, or self-presentation effects may influence what is remembered or emphasized.	Objective capture of speech, behavior, and performance indicators, allowing astronauts to review what was said, when it was said, and how it related to operational events.	It could improve evidence-based reflection and support metacognitive calibration by comparing perceived performance with recorded task performance.	After a high-workload simulation, astronauts could compare their recalled decision-making process with time-stamped communication and task data to identify missed invariants or successful adaptations.
Facilitator variability	Debriefing quality may depend on the expertise of the facilitator, which can introduce variability between instructors and limit scalability.	Structured prompting through LLM-based reflective dialogue, using context-sensitive questions to support self-explanation and discussion.	It could provide a more consistent structure for reflection while preserving the role of human facilitators in interpretation and judgment.	In astronaut training, AI could suggest standardized reflection prompts, while the instructor remains responsible for guiding the discussion and validating the interpretation.
Limited continuity across missions or training sessions	Manual documentation may make it difficult to trace the evolution of skills, procedures, or recurring performance patterns across multiple sessions.	Archiving, indexing, and pattern detection across debriefing records and repeated training events.	It could help identify recurrent cognitive or behavioral tendencies and supports individualized learning plans or targeted exercises.	Across repeated EVA simulations, AI could identify recurring communication delays, hesitation during high-workload phases, or repeated procedural bottlenecks.
Need for autonomous reflection in deep-space missions	During long-duration or Mars missions, communication delays may prevent real-time feedback from Earth-based instructors, requiring crews to perform more autonomous self-debriefing.	AI-assisted self-debriefing, including structured summaries, adaptive prompts, and chatbot-like reflective support.	It could support self-regulated learning when immediate ground support is unavailable, helping crews maintain reflection and adaptation during isolated operations.	During a Mars analog mission or future deep-space operation, AI could guide a crew through a post-task self-debrief when communication with Earth is delayed.
Maintaining ecological validity	Astronaut training must preserve the perceptual, cognitive, and environmental demands of real operations.	Integrated feedback displays combining helmet-camera footage, telemetry, physiological signals, and communication data.	It could help astronauts reconstruct perception-action couplings and better understand how environmental information, movement, and communication shaped performance.	In EVA simulation, AI could help visualize how movement trajectories, communication latency, and physiological workload evolved during critical task phases.
Ethical, privacy, and trust concerns	AI-supported debriefing may involve sensitive spoken, behavioral, or performance data, and the presence of an evaluative system may affect openness and psychological safety.	Privacy-preserving design, anonymization or partial redaction, transparent data handling, access control, and human oversight.	It could protect trust, psychological safety, and the interpretability of debriefing data while ensuring that AI output remains subject to human judgment.	AI-generated debriefing summaries could remove personally identifiable details before archiving, while instructors retain responsibility for final interpretation and feedback.

precision and recall of detected events or affective markers, user trust, and the perceived usefulness of AI-generated outputs.

Finally, the implications of AI-supported debriefing may extend beyond the field of astronautics [4]. Several of the concepts discussed in this paper, such as facilitating reflection, supporting self-regulated learning, and integrating multiple sources of performance data, are also relevant in other high-performance domains. Similar approaches could be considered in areas such as healthcare, education, emergency response, and other operational environments in which individuals and teams must learn from complex experiences. As has often occurred with research conducted in microgravity aboard the ISS and later transferred to terrestrial applications, astronaut debriefing may provide a valuable testing ground for exploring how AI may support learning, performance improvement, and professional development in real-world settings.

6. Conclusion

This perspective paper examined an ecological and learning-oriented framework for the integration of AI into astronaut debriefing. Debriefing is not only a mechanism for error analysis but a process for recalibrating perception–action couplings and fostering adaptive expertise in high-stakes environments like aviation and astronautics. AI applications, especially LLMs and speech-to-text systems, could enhance this process by automating documentation, structuring reflection, and supporting both instruc-

tors and astronauts in deriving insights from performance data. Hence, AI tools could potentially be integrated into debriefing processes not only for training purposes but also within operational contexts, such as those for the long-duration missions, supporting self-regulated learning even when crews are isolated from Earth. Yet, the paper reviewed that future research should first establish the validity and reliability of these technologies before they can be adopted empirically in operational settings. Furthermore, their use must adhere to ethical principles, safeguard privacy, and preserve human judgment as the foundation of reflective practice. In conclusion, the integration of AI into astronaut debriefing could be a significant step toward the human–AI collaboration for trainings and learning within extreme environments, combining computational intelligence with ecological understanding to advance human performance and training effectiveness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Simone Caso: Writing – original draft, Investigation, Conceptualization. **Yannick Hill:** Writing – review & editing, Writing – original draft.

References

[1] R.M. Fanning, D.M. Gaba, The role of debriefing in simulation-based learning, Simul. Healthc. 2 (2007) 115–125.
[2] N.L. Keiser, W. Arthur Jr., A meta-analysis of the effectiveness of the after-action review (or debrief) and factors that influence its effectiveness, J. Appl. Psychol. 106 (2021) 1007–1032.
[3] S. Tannenbaum, C.P. Cerasoli, Do team and individual debriefs enhance performance? A meta-analysis, Hum. Fact. 55 (2013) 231–245, doi:10.1177/0018720812448394.
[4] S. Caso, Y. Hill, T.J. Mavin, Enhancing debriefing practices through artificial intelligence: an ecological and cross-domain perspective, Int. J. Train. Res. (2026) 1–16, doi:10.1080/14480220.2026.2708657.
[5] S. Buckle, R. Peldszus, L. Bessone, Adaptation of the ISS Human behaviour & performance competency model as observation & debriefing tool for mission control teams during simulations, in: T. Sgobba, I. Rongier (Eds.), Space Safety is No Accident, Springer, Cham, 2015, doi:10.1007/978-3-319-15982-9_36.
[6] D.L. Dempsey, I. Barshi, Applying research-based training principles: toward crew-centered, mission-oriented space flight training, in: Psychology and Human Performance in Space Programs, CRC Press, 2020, pp. 63–80.
[7] S. Caso, Training adaptable astronauts: integrating ecological psychology into human performance in space missions, Ecol. Psychol. 38 (2026) 73–88, doi:10.1080/10407413.2025.2582871.
[8] S.I. Dev, A.M. Khader, S.R. Begerowski, S.R. Anderson, G. Clément, S.T. Bell, Cognitive performance in ISS astronauts on 6-month low earth orbit missions, Front. Physiol. 15 (2024) 1451269, doi:10.3389/fphys.2024.1451269.
[9] I.K. Garaszczuk, K. Komorowska, M.A. Rusnak, The impact of space habitat conditions on visual performance and cognitive load in analogue astronauts, Acta Astronaut. 228 (2025) 664–674, doi:10.1016/j.actaastro.2024.12.046.
[10] F. Pagnini, D. Manzey, E. Rosnet, D. Ferravante, O. White, N. Smith, Human behavior and performance in deep space exploration: next challenges and research gaps, NPJ Micrograv. 9 (2023) 27, doi:10.1038/s41526-023-00270-7.
[11] M. Wilson, Six views of embodied cognition, Psychon. Bull. Rev. 9 (2002) 625–636.
[12] J.H. Flavell, Metacognition and cognitive monitoring: a new area of cognitive–developmental inquiry, Am. Psychol. 34 (1979) 906–911, doi:10.1037/0003-066X.34.10.906.
[13] S. Caso, Perceiving and acting on other worlds, Ecol. Psychol. (2026) 1–18, doi:10.1080/10407413.2026.2713442.
[14] Y. Kikkawa, T.J. Mavin, A review of debriefing practices: toward a framework for airline pilot debriefing, Aviat. Psychol. Appl. Hum. Fact. 7 (2017) 42–54, doi:10.1027/2192-0923/a000114.
[15] T.J. Mavin, Y. Kikkawa, S. Billett, Key contributing factors to learning through debriefings: commercial aviation pilots' perspectives, Int. J. Train. Res. 16 (2018) 122–144, doi:10.1080/14480220.2018.1501906.
[16] J. Vaughn, S.H. Ford, A. Sheckells, H. Fasnacht, D. Crawford, G. Dogan, Enhancing simulation facilitator debriefing using a generative artificial intelligence feedback interface and retrieval-augmented generation: a pilot study, Clin. Simul. Nurs. 109 (2025) 101843, doi:10.1016/j.ecns.2025.101843.
[17] A. Chemero, An outline of a theory of affordances, Ecol. Psychol. 15 (2003) 181–195, doi:10.1207/S15326969ECO1502_5.
[18] B.R. Fajen, M.A. Riley, M.T. Turvey, Information, affordances, and the control of action in sport, Int. J. Sport Psychol. 40 (2009) 79–107.
[19] J.J. Gibson, The Senses Considered as Perceptual Systems, Houghton Mifflin, 1966.
[20] J.J. Gibson, The Ecological Approach to Visual Perception: Classic Edition, 1st ed., Psychology Press, 1979, doi:10.4324/9781315740218.
[21] H. Heft, An ecological approach to psychology, Rev. Gen. Psychol. 17 (2013) 162–167, doi:10.1037/a0032928.
[22] D.M. Jacobs, C.F. Michaels, Direct learning, Ecol. Psychol. 19 (2007) 321–349, doi:10.1080/10407410701432337.
[23] T.A. Stoffregen, Affordances as properties of the animal-environment system, Ecol. Psychol. 15 (2003) 115–134.
[24] O. Bensch, L. Bensch, T. Nilsson, F. Saling, W.M. Sadri, C. Hartmann, T. Hecking, J.N. Kutz, Towards a Reliable Offline Personal AI Assistant For Long Duration Spaceflight, 2024, doi:10.48550/arXiv.2410.16397.
[25] M. Henquet, T. Bellucci, AI-assisted debrief: automated flight debriefing summarization and competency assessment, in: Proceedings of the 2nd International Conference on Cognitive Aircraft Systems (ICCAS), 1, SciTePress, 2024, pp. 72–79, doi:10.5220/0012970200004562.
[26] L. Gonzalez, A. Nagendran, Artificial intelligence (AI)-facilitated debriefing: a pilot study, Clin. Simul. Nurs. 105 (2025) 101782, doi:10.1016/j.ecns.2025.101782.
[27] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Gunnemann, E. Hullermeier, S. Krusche, G. Kutyniok, T. Michaeli, C. Nerdel, J. Pfeffer, O. Poquet, M. Sailer, A. Schmidt, T. Seidel, M. Stadler, J. Weller, J. Kuhn, G. Kasneci, ChatGPT for good? On opportunities and challenges of large language models for education, (2023). https://doi.org/10.35542/osf.io/5er8f.
[28] D. Myers, R. Mohawesh, V.I. Chellaboina, A.L. Sathvik, P. Venkatesh, Y.-H. Ho, H. Henshaw, M. Alhawawreh, D. Berdik, Y. Jararweh, Foundation and large language models: fundamentals, challenges, opportunities, and social impacts, Clust. Comput. 27 (2024) 1–26, doi:10.1007/s10586-023-04203-7.
[29] L. Floridi, M. Chiriatti, GPT-3: its nature, scope, limits, and consequences, Minds Mach. 30 (2020) 681–694, doi:10.1007/s11023-020-09548-1.
[30] R. Huijzer, Y. Hill, Large language models show human behavior, (2023). https://doi.org/10.31234/osf.io/munc9.
[31] T. Marler, L.L. Rohn, M. Toukan, G. Mazzotta, S. Young, E. Ro, M. Sargent, M. Strawn, M. Rodgers, U.S. Space Force Operational Test and Training Infrastructure (OTTI): Recommendations for Organizational Roles, Responsibilities, and Authorities, RAND Corporation, Santa Monica, CA, 2025.
[32] T. Shmelova, A. Sterenharz, S. Dolgikh, Artificial intelligence in aviation industries: methodologies, education, applications, and opportunities, in: T. Shmelova, Y. Sikirda, A. Sterenharz (Eds.), Handbook of Research on Artificial Intelligence Applications in the Aviation and Aerospace Industries, IGI Global, 2020, pp. 1–35, doi:10.4018/978-1-7998-1415-3.ch001.
[33] A. Bandura, Self-efficacy mechanism in human agency, Am. Psychol. 37 (1982) 122–147.
[34] A. Bandura, Self-Efficacy: The exercise of Control, W.H. Freeman, New York, 1997.
[35] A. Bandura, S. Wessels, Self-Efficacy, Cambridge University Press, Cambridge, 1997.
[36] B.J. Zimmerman, Attaining self-regulation: a social cognitive perspective, in: M. Boekaerts, P.R. Pintrich, M. Zeidner (Eds.), Handbook of Self-Regulation, Academic Press, 2000, pp. 13–39, doi:10.1016/B978-012109890-2/50031-7.
[37] W.C. Phinney, Science Training History of the Apollo Astronauts, Johnson Space Center, Houston, TX, 2015 NASA/SP-2015-626 https://ntrs.nasa.gov/citations/20190026783 .
[38] P.E. Clark, Extreme Exploration: Celebrating and Applying Lessons Learned from the Apollo Expeditions, Springer Nature, Cham, 2025.
[39] L.P. Halamek, Lost Moon, saved Lives: using the movie Apollo 13 as a video primer in behavioral skills for simulation trainees and instructors, Simul. Healthc. 5 (5) (2010) 303–310, doi:10.1097/SIH.0b013e3181e5e329.
[40] R.C. Kohler, L. Reeder, Mission Training Program For the Apollo Lunar Landing Mission, NASA Technical Memorandum NASA-TM-X-65078, 1968.
[41] J.J. Van Bockel, Mission Training Program For Third Manned Apollo Mission (No. MSC-IN-CF-D-67-12), 1967.
[42] P.E. Clark, Overview: the Apollo lunar surface expeditionary approach, Extreme Exploration, Springer, 2025, doi:10.1007/978-3-032-01734-5_1.
[43] G.W. Hoffler, R.A. Wolthuis, R.L. Johnson, Apollo space crew evaluations, Aerosp. Med. (1974) 807.
[44] P.C. Kramer, Apollo Experience Report: Systems and Flight Procedures Development, 1973 NASA-TN-D-7436.
[45] C.H. Woodling, S. Faber, J.J. Vanbockel, C.C. Olasky, W.K. Williams, J.L. Mire, J.R. Homer, Apollo Experience Report: Simulation of Manned Space Flight For Crew Training, 1973 MSC-S-346.
[46] National Aeronautics and Space Administration Shuttle Mission Crew Debriefing Report: Analysis and Recommendations, 1997 NASA Technical Memorandum 19970015346.
[47] L. Martin, J. Orasanu, E. Villeda, Instructor Debrief Training in SPOT, 2002 NASA Technical Memorandum 20020068947.
[48] C.W. Nielsen, A.V. Østerballe, R.L. Tougaard, J. Struble, R.G. Larsen, B. Lohmann, P. Jaekel, T. Weber, M. de Zee, E. De Martino, Kinematic analysis of the European Enhanced Exploration Exercise Device in unpowered mode, NPJ Micrograv. 12 (2026) 41, doi:10.1038/s41526-026-00588-y.

[49] L.B. Landon, L.G. Morissette, S.T. Bell, Private astronaut post-mission debrief interviews study, Human Research Program Investigators' Workshop (HRP IWS), NASA Technical Reports Server, 2025.
[50] G.W. Liddell, Human factors application of long duration mission post-flight debriefs: international Space Station phase I, SAE Technical Paper 981785, 1998.
[51] P.A. Santy, A.W. Holland, L. Looper, R. Marcondes-North, Results of an international space crew debrief, Aerospace Medical Association 63rd Annual Scientific Meeting Program, 1992.
[52] L.P. Halamek, R.A. Cady, M.R. Sterling, Using briefing, simulation and debriefing to improve human and system performance, Semin. Perinatol. 43 (2019) 151178.
[53] E. Hong, S. Kazmir, B. Dylik, M. Auerbach, M. Rosati, S. Athanasopoulou, R. Himmelstein, T.M. Whitfill, L. Johnston, T.A. Wolbrink, A.S. Rosen, I.T. Gross, Exploring the use of a large language model in simulation debriefing: an observational simulation-based pilot study, Simul. Healthc. 20 (2025) 366–371, doi:10.1097/SIH.0000000000000861.
[54] V.M. Reddy, T. Vaishnavi, K.P. Kumar, Speech-to-text and text-to-speech recognition using deep learning, in: 2nd International Conference on Edge Computing and Applications (ICECAA), 2023, pp. 657–666, doi:10.1109/ICECAA58104.2023.10212222.
[55] M. Benfatah, I. Youlyouz-Marfak, E. Saad, A. Hilali, C. Nejjari, A. Marfak, Impact of artificial intelligence-enhanced debriefing on clinical skills development in nursing students: a comparative study, Teach. Learn. Nurs. 19 (2024) e574–e579, doi:10.1016/j.teln.2024.04.007.
[56] H. Mohapatra, S.R. Mishra, Unlocking insights: exploring data analytics and AI tool performance across industries, in: P. Singh, A.R. Mishra, P. Garg (Eds.), Data Analytics and Machine Learning, Stud. Big Data, Data Analytics and Machine Learning, Stud. Big Data, 145, Springer, Singapore, 2024, doi:10.1007/978-981-97-0448-4_13.
[57] S. Caso, Emerging technologies in military space operations: current applications and future research for educational and training purposes, Int. J. Train. Res. 23 (2025) 140–155, doi:10.1080/14480220.2024.2431482.
[58] S.E. Kase, C.P. Hung, T. Krayzman, J.Z. Hare, B.C. Rinderspacher, S.M. Su, The future of collaborative human-artificial intelligence decision-making for mission planning, Front. Psychol. 13 (2022) 850628, doi:10.3389/fpsyg.2022.850628.
[59] P. Baglioni, R. Fisackerly, B. Gardini, G. Gianfiglio, A.L. Pradier, A. Santovincenzo, J.L. Vago, M. van Winnendael, The Mars exploration plans of ESA, IEEe Robot. Autom. Mag. 13 (2006) 83–89, doi:10.1109/MRA.2006.1638019.
[60] S. Creech, J. Guidi, D. Elburn, Artemis: an overview of NASA's activities to return humans to the Moon, in: 2022 IEEE Aerospace Conference (AERO), 2022, pp. 1–7, doi:10.1109/AERO53065.2022.9843277.
[61] S.K. Johnson, D.J. Mortensen, M.A. Chavez, C.L. Woodland, Gateway – a communications platform for lunar exploration, in: Proc. 38th Int. Commun. Satell. Syst. Conf. (ICSSC 2021), 2022, pp. 9–16, doi:10.1049/icp.2022.0544.
[62] C. Caini, R. Firrincieli, T. de Cola, I. Bisio, M. Cello, G. Acar, Mars to Earth communications through orbiters: delay-tolerant/disruption-tolerant networking performance analysis, Int. J. Satell. Commun. Netw. 32 (2014) 127–140, doi:10.1002/sat.1043.
[63] S.P. Yamaguchi, A.Mora Vargas, T. Eisenberg, C. Rogon, T. Yamamoto, S. Inoue, C. Kössl, B. Coltin, T. Smith, J.V. Benavides, Free-flying crew cooperative robots on the ISS: a joint review of Astrobee, CIMON, and Int-Ball operations, (2026). https://doi.org/10.48550/arXiv.2602.10686.
[64] G. Bansal, V. Chamola, A. Hussain, M. Guizani, D. Niyato, Transforming conversations with AI—a comprehensive study of ChatGPT, Cogn. Comput. 16 (2024) 2487–2510, doi:10.1007/s12559-023-10236-2.
[65] P.A. Santy, R.T. Jennings, Human reproductive issues in space, Adv. Space Res. 12 (1992) 151–155.
[66] A. Edmondson, Psychological safety and learning behavior in work teams, Adm. Sci. Q. 44 (1999) 350–383, doi:10.2307/2666999.
[67] Y.Y. Wang, Y.W. Chuang, Artificial intelligence self-efficacy: scale development and validation, Educ. Inf. Technol. 29 (2024) 4785–4808, doi:10.1007/s10639-023-12015-w.
[68] S. Caso, D. Vlaskamp, A. Landman, Applying sport psychology to aviation: ensuring resilient pilot performance through self-regulation techniques and tools, Aviat. Psychol. Appl. Hum. Fact. 16 (1) (2026) 42–54, doi:10.1027/2192-0923/a000315.
[69] N. Kanas, D. Manzey, Basic issues of human adaptation to space flight, Space Psychology and Psychiatry, 22, Springer, Dordrecht, 2008, doi:10.1007/978-1-4020-6770-9_2.
[70] S. Caso, P. Furley, G. Jordet, Using video-notational analysis to examine soccer players' behaviours, Int. J. Sport Exerc. Psychol. 24 (2026) 739–759, doi:10.1080/1612197X.2025.2477165.
[71] S. Caso, Applying sport and performance psychology to space exploration: enhancing astronaut training, performance and mission readiness, Int. J. Sport Exerc. Psychol. (2026) 1–20 Advance online publication, doi:10.1080/1612197X.2026.2730159.
[72] S. Grinschgl, G. Hofer, A.C. Neubauer, Improving self-knowledge: how performance feedback impacts individuals' self-estimates of their cognitive abilities, J. Res. Pers. 116 (2025) 104606, doi:10.1016/j.jrp.2025.104606.
[73] J.H. Nieminen, D. Boud, Student self-assessment: a meta-review of five decades of research, Assess. Educ. 32 (2025) 127–151, doi:10.1080/0969594X.2025.2510211.
[74] K.A. Ericsson, The influence of experience and deliberate practice on the development of superior expert performance, in: The Cambridge Handbook of Expertise and Expert Performance, 2006, pp. 685–705.
[75] G.-J. Hwang, H. Xie, B.W. Wah, D. Gašević, Vision, challenges, roles and research issues of artificial intelligence in education, Comput. Educ. Artif. Intell. 1 (2020) 100001, doi:10.1016/j.caeai.2020.100001.
[76] C. Jeffs, N. Nelson, K.A. Grant, L. Nowell, B. Paris, N. Viceer, Feedback for teaching development: moving from a fixed to growth mindset, Prof. Dev. Educ. 49 (2023) 842–855, doi:10.1080/19415257.2021.1876149.
[77] E.J. de Visser, R. Pak, T.H. Shaw, From 'automation' to 'autonomy': the importance of trust repair in human–machine interaction, Ergonomics 61 (2018) 1409–1427, doi:10.1080/00140139.2018.1457725.
[78] J.D. Lee, K.A. See, Trust in automation: designing for appropriate reliance, Hum. Fact. 46 (2004) 50–80, doi:10.1518/hfes.46.1.50_30392.
[79] A.K.Y. Yanamala, S. Suryadevara, Advances in data protection and artificial intelligence: trends and challenges, Int. J. Adv. Eng. Technol. Innov. 1 (2023) 294–319.
[80] A.K.Y. Yanamala, S. Suryadevara, Navigating data protection challenges in the era of artificial intelligence: a comprehensive review, Rev. Intell. Artif. Med. 15 (2024) 113–146.
[81] A. Shukla, Utilizing AI and machine learning for human emotional analysis through speech-to-text engine data conversion, J. Artif. Intell. Cloud Comput. 1 (2022) 1, doi:10.47363/JAICC/2022(1)145.
[82] L. Gerlach, Anonymization and Identifiability: Enhancing Data Protection through Differential Privacy and Artificial Intelligence, 8, Walter de Gruyter GmbH & Co. KG, 2025.
[83] L. Yang, M. Tian, D. Xin, Q. Cheng, J. Zheng, AI-driven anonymization: protecting personal data privacy while leveraging machine learning, (2024).
[84] H. Terai, S. Kai, M. Oi, Effects of anonymity on decision making and evaluations of others' decisions in moral dilemmas, Acta Psychol. 259 (2025) 105325, doi:10.1016/j.actpsy.2025.105325.
[85] B. Ivan, O. Olga, O. Oksana, R. Roman, M. Marta, K. Khrystyna, Application of grey relational analysis for utilizing artificial intelligence methods in aviation management, in: R. Khamis, A. Buallay (Eds.), AI in Business: Opportunities and Limitations, AI in Business: Opportunities and Limitations, 515, Springer Nature Switzerland, 2024, pp. 113–123, doi:10.1007/978-3-031-48479-7_11.
[86] I. Shafi, M.F. Mazhar, A. Fatima, R.M. Alvarez, Y. Miró, J.C.M. Espinosa, I. Ashraf, Deep learning-based real time defect detection for optimization of aircraft manufacturing and control performance, Drones 7 (2023) 31, doi:10.3390/drones7010031.
[87] P.A. Abdalla, A. Varol, Advantages to disadvantages of cloud computing for small-sized business, in: 2019 7th International Symposium on Digital Forensics and Security (ISDFS), 2019, pp. 1–6, doi:10.1109/ISDFS.2019.8757549.
[88] N. Pramod, A.K. Muppalla, K.G. Srinivasa, Limitations and challenges in cloud-based applications development, in: Z. Mahmood, S. Saeed (Eds.), Software Engineering Frameworks For the Cloud Computing Paradigm, Springer, London, 2013, pp. 55–75, doi:10.1007/978-1-4471-5031-2_3.
[89] E. Kulida, V. Lebedev, Problems in applying machine learning methods to aviation, in: 16th International Conference Management of Large-Scale System Development (MLSD), 2023, pp. 1–4, doi:10.1109/MLSD58227.2023.10304039.
[90] G.G. De la Torre, G. Groemer, A. Diaz-Artiles, N. Pattyn, J. Van Cutsem, M. Musilova, W. Kopec, S. Schneider, V. Abeln, T. Larose, F. Ferlazzo, P. Zivi, A. de Carvalho, G.M. Sandal, L. Orzechowski, M. Nicolas, R. Billette de Villemeur, A.P. Traon, I. Antunes, Space analogs and behavioral health performance research review and recommendations checklist from ESA Topical Team, NPJ Micrograv. 10 (2024) 98, doi:10.1038/s41526-024-00437-w.
[91] K.K. Schlosser, I. Cinelli, T. Waelde, L.L. Álvarez, G. Pokorádi, K. Pósch, I. Whiteley, Optimise behavioural health and human factors research for deep space missions by classifying analogue scenarios and fidelity, Front. Space Technol. 6 (2025), doi:10.3389/frspt.2025.1391331.
[92] D. Araujo, K. Davids, Team synergies in sport: theory and measures, Front. Psychol. 7 (2016) 1449, doi:10.3389/fpsyg.2016.01449.
[93] D. Araújo, K. Davids, P. Passos, Ecological validity, representative design, and correspondence between experimental task constraints and behavioral setting: comment on Rogers, Kadar, and Costall (2005), Ecol. Psychol. 19 (2007) 69–78, doi:10.1080/10407410709336951.
[94] E. Brunswik, Perception and the Representative Design of Psychological Experiments, University of California Press, 1956.
[95] R.A. Pinder, K. Davids, I. Renshaw, D. Araújo, Representative learning design and functionality of research and practice in sport, J. Sport Exerc. Psychol. 33 (2011) 146–155, doi:10.1123/jsep.33.1.146.
[96] J. Rasmussen, A.M. Pejtersen, L.P. Goodstein, Cognitive Systems Engineering, Wiley, 1994.
[97] K.J. Vicente, Cognitive Work Analysis: Toward Safe, Productive, and Healthy Computer-Based Work, Lawrence Erlbaum Associates, 1999.
[98] K.J. Vicente, J. Rasmussen, Ecological interface design: theoretical foundations, IEEE Trans. Syst. Man Cybern. 22 (1992) 589–606, doi:10.1109/21.156574.
[99] E.A. Locke, E. Frederick, C. Lee, P. Bobko, Effect of self-efficacy, goals, and task strategies on task performance, J. Appl. Psychol. 69 (2) (1984) 241–251, doi:10.1037/0021-9010.69.2.241.